Abstract Algebra III: Assignment - Week 12

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Problem 1

(i) Let (X, ρ) be a transitive **Set**-representation of G. For $x \in X$, let G_x denote the fixator of x in G. Then the map

$$G/G_x \to X, \quad gG_x \mapsto \rho(g)(x)$$

defines an isomorphism $(G/G_x, \rho_{G_x}) \xrightarrow{\sim} (X, \rho)$.

(ii) For subgroups H and H' of G, the representations $(G/H, \rho_H)$ and $(G/H', \rho_{H'})$ are isomorphic if and only if H and H' are conjugate in G.

Solution: (i) Let $f : G/G_x \to X$ denote the map we discuss. First we show that f is well-defined. Suppose

$$gG_x = hG_x.$$

Then $h^{-1}g \in G_x$, so

$$\rho_{h^{-1}g}(x) = x.$$

Hence

$$\rho_g(x) = \rho_h(\rho_{h^{-1}g}(x)) = \rho_h(x).$$

Therefore $f(gG_x) = f(hG_x)$, so f is well-defined. Next, We show that f is a bijection of sets. Let y be an arbitrary element of X that distinct to x. Since G act transitively on X then there exists $g \in G$ such that $\rho_g(x) = y$. Therefore, we have

$$f(gG_x) = \rho_g(x) = y.$$

It implies that f is surjective. Let $g, h \in G$ such that $f(gG_x) = f(hG_x)$. It implies that $\rho_g(x) = \rho_h(x)$ and thus

$$x = \rho_{h^{-1}} \circ \rho_g(x) = \rho_{h^{-1} \cdot g}(x).$$

Then we have $h^{-1} \cdot g \in G_x$. Equivalently, we have $gG_x = hG_x$. Therefore, f is injective. For arbitrary $g \in G$, we then have

$$f(g \cdot hG_x) = f(ghG_x) = \rho_{gh}(x) = \rho_g \circ \rho_h(x) = \rho_g \cdot f(hG_x).$$

Therefore, f is isomorphism of representation.

(ii) Suppose first that

$$f : G/H \to G/H'$$

is an isomorphism of G-sets. Write

$$f(H) = gH'$$

for some $g \in G$. Since f is G-equivariant, it preserves stabilizers. Hence

$$H = \mathrm{Stab}_G(H) = \mathrm{Stab}_G(f(H)) = \mathrm{Stab}_G(gH') = gH'g^{-1}.$$

Thus H and H' are conjugate.

Conversely, suppose

$$H = gH'g^{-1}$$

for some $g \in G$. Define

$$\Phi : G/H \to G/H', \qquad \Phi(aH) = agH'.$$

This is well-defined: if $aH = bH$, then $a = bh$ for some $h \in H$. Since

$$g^{-1}hg \in H',$$

we get

$$agH' = bhgH' = bgH'.$$

Thus $\Phi(aH) = \Phi(bH)$.

The map Φ is bijective, with inverse

$$\Psi : G/H' \to G/H, \qquad \Psi(aH') = ag^{-1}H.$$

Moreover, for $k \in G$,

$$\Phi(k \cdot aH) = \Phi(kaH) = kagH' = k \cdot \Phi(aH).$$

Therefore Φ is G-equivariant. Hence

$$(G/H, \rho_H) \cong (G/H', \rho_{H'})$$

if and only if H and H' are conjugate in G. ∎

Problem 2

Assume that G is the cyclic (additive) group $\mathbb{Z}/p\mathbb{Z}$ where p is a prime number, and let k be a field of characteristic p. Consider the morphism

$$G \to \mathrm{GL}_2(k), \quad \lambda \mapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}.$$

Prove that the line generated by the first standard basis vector of k^2 is G-stable but has no G-stable complement.

Solution: Let
$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$
Since $\mathrm{char}(k) = p$, the map
$$\rho : \mathbb{Z}/p\mathbb{Z} \to \mathrm{GL}_2(k), \qquad \lambda \mapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}$$
is well-defined and is a group homomorphism.

Let
$$W = ke_1.$$
Then for every $\lambda \in \mathbb{Z}/p\mathbb{Z}$,
$$\rho(\lambda)e_1 = e_1,$$
so W is G-stable.

We claim that W has no G-stable complement. Suppose, for contradiction, that
$$k^2 = W \oplus U$$
with U is G-stable. Then U is one-dimensional and $U \neq W$, so
$$U = k(ae_1 + e_2)$$
for some $a \in k$. Since U is G-stable, it is stable under the action of $1 \in \mathbb{Z}/p\mathbb{Z}$. Thus
$$\rho(1)(ae_1 + e_2) = (a+1)e_1 + e_2$$
must lie in U. Hence there exists $c \in k$ such that
$$(a+1)e_1 + e_2 = c(ae_1 + e_2).$$
Comparing coefficients gives $c = 1$, and then
$$a + 1 = a,$$
a contradiction. Therefore W has no G-stable complement. ■

Problem 3

Let (M, ρ) and (N, σ) be two k-representations of G. Fix basis $(e_i)_{1 \le i \le m}$ and $(f_j)_{1 \le j \le n}$ of M and N respectively. Then $(e_i \otimes f_j)_{1 \le i \le m, 1 \le j \le n}$ is a basis of $M \otimes_k N$. Denote by (m, ρ), (n, σ), $(mn, \rho \otimes \sigma)$ be the matrix representations associated with (M, ρ), (N, σ), $(M \otimes_k N, \rho \otimes \sigma)$ respectively. Determine the relation between $(\rho \otimes \sigma)(g)$ and $\rho(g)$, $\sigma(g)$.

Solution: Suppose
$$\rho_g(e_i) = \sum_{k=1}^{m} a_{ki} e_k, \qquad \sigma_g(f_j) = \sum_{\ell=1}^{n} b_{\ell j} f_\ell.$$

Then

$$(\rho \otimes \sigma)_g(e_i \otimes f_j) = \rho_g(e_i) \otimes \sigma_g(f_j) = \sum_{k=1}^{m} \sum_{\ell=1}^{n} a_{ki} b_{\ell j} (e_k \otimes f_\ell).$$

With respect to the ordered basis

$$(e_1 \otimes f_1, \ldots, e_1 \otimes f_n, e_2 \otimes f_1, \ldots, e_m \otimes f_n),$$

the column corresponding to $e_i \otimes f_j$ is

$$(a_{1i}b_{1j}, \ldots, a_{1i}b_{nj}, a_{2i}b_{1j}, \ldots, a_{2i}b_{nj}, \ldots, a_{mi}b_{1j}, \ldots, a_{mi}b_{nj})^T.$$

Therefore

$$[(\rho \otimes \sigma)_g] = [\rho_g] \otimes [\sigma_g],$$

where the tensor product on the right is the Kronecker product of matrices. ■

Problem 4

Let (M, ρ) be a k-representations of G. Fix basis $(e_i)_{1 \le i \le m}$ of M. Let (M^*, ρ^*) be the contragredient representation of (M, ρ). Denote by (m, ρ), (m, ρ^*) be the matrix representations associated with (M, ρ), (M^*, ρ^*). Determine the relation between $\rho(g)$ and $\rho^*(g)$.

Solution: Let $e_1, \ldots, e_n$ be a basis of V, with dual basis $e_1^*, \ldots, e_n^*$. Suppose

$$\rho_g(e_i) = \sum_{k=1}^{n} a_{ki} e_k.$$

Thus

$$[\rho_g] = (a_{ki}).$$

The contragredient representation ρ^* is defined by

$$(\rho_g^* \varphi)(v) = \varphi(\rho_{g^{-1}} v).$$

Write

$$\rho_{g^{-1}}(e_j) = \sum_{k=1}^{n} b_{kj} e_k.$$

Then $[\rho_{g^{-1}}] = (b_{kj}) = [\rho_g]^{-1}$. For each i, j,

$$(\rho_g^* e_i^*)(e_j) = e_i^*(\rho_{g^{-1}} e_j) = e_i^* \left(\sum_{k=1}^{n} b_{kj} e_k \right) = b_{ij}.$$

Hence the coefficient of e_j^* in $\rho_g^*(e_i^*)$ is b_{ij}. Therefore

$$[\rho_g^*] = (b_{ij}) = [\rho_{g^{-1}}]^T = ([\rho_g]^{-1})^T.$$

Thus the matrix of the contragredient representation is the inverse transpose of the original matrix. ■

Problem 5

(i) Let G be a finite group acting on a finite set Ω. Let $\mathrm{Fix}^G(\Omega)$ denote the set of fixed points of Ω under G, and let $\Omega^\#$ denote the set of orbits of G on $\Omega \setminus \mathrm{Fix}^G(\Omega)$. Prove that

$$|\Omega| = |\mathrm{Fix}^G(\Omega)| + \sum_{\omega\in\Omega^\#} |\omega|.$$

From now on, we assume that p is a prime number and that G is a nontrivial p-group.

(ii) Assume now that $|\Omega|$ is a power of p different from 1. Prove that $|\mathrm{Fix}^G(\Omega)|$ is divisible by p.

(iii) Let k be a (commutative) field of characteristic p. Let M be a kG-module. Prove that $\mathrm{Fix}^G(M) \neq 0$. (Hint. (a) Let $(e_i)_{i=1,\dots,r}$ be a k-basis of M. Prove that the abelian group generated by $(ge_i)_{i=1,\dots,r\ (g\in G)}$ is an $\mathbb{F}_pG$-module. (b) Apply question (2) above to conclude.)

(iv) Prove that, up to isomorphism, the trivial representation of G is the unique irreducible kG-module.

Solution: (i) The set Ω decomposes as a disjoint union

$$\Omega = \mathrm{Fix}^G(\Omega) \sqcup \big(\Omega \setminus \mathrm{Fix}^G(\Omega)\big).$$

Moreover, $\Omega \setminus \mathrm{Fix}^G(\Omega)$ is G-stable, hence it is a disjoint union of its G-orbits:

$$\Omega \setminus \mathrm{Fix}^G(\Omega) = \bigsqcup_{\omega\in\Omega^\#} \omega.$$

Therefore

$$\Omega = \mathrm{Fix}^G(\Omega) \sqcup \bigsqcup_{\omega\in\Omega^\#} \omega.$$

Taking cardinalities gives

$$|\Omega| = |\,\mathrm{Fix}^G(\Omega)| + \sum_{\omega\in\Omega^\#} |\omega|.$$

(ii) We just prove that for any $\omega \in \Omega^\#$ we have $|\omega|$ is divisible by p. Consider ω as transitive G-set, according to (1), we choose $x \in \omega$ then it induces the bijection

$$G/G_x \xrightarrow{\sim} \omega.$$

Since the only prime divisor of $|G|$ is p, we can suppsoe that $|\omega| = |G/G_x| = p^s$. If $s = 0$, it turns out that $x \in \mathrm{Fix}^G(\Omega)$, which contradicts to our assumption. Therefore, $|\Omega|$ and each $|\omega|$ is divisible by p implies that

$$|\,\mathrm{Fix}^G(\Omega)| = |\Omega| - \sum_{\omega\in\Omega^\#} |\omega|.$$

is also divisible by p.

(iii) Assume $M \neq 0$, and let $e_1, \ldots, e_r$ be a k-basis of M. Set

$$M' = \langle ge_i \mid g \in G,\ 1 \leq i \leq r\rangle_{\mathbb{F}_p} \subset M.$$

Since $\mathrm{char}(k) = p$, this is a finite $\mathbb{F}_p$-subspace of M. It is G-stable, because for $h \in G$,

$$h(ge_i) = (hg)e_i \in M'.$$

Thus G acts on the finite set M'.

Since $M' \neq 0$, we have $|M'| = p^n$ for some $n \geq 1$. By part (ii),

$$|\operatorname{Fix}^G(M')|$$

is divisible by p. But $0 \in \operatorname{Fix}^G(M')$, so the fixed point set cannot consist only of 0. Hence there exists

$$0 \neq m \in \operatorname{Fix}^G(M').$$

Since $M' \subset M$, we get

$$0 \neq m \in \operatorname{Fix}^G(M).$$

Therefore

$$\operatorname{Fix}^G(M) \neq 0.$$

(iv)Let M be an irreducible kG-module. By part (iii),

$$\operatorname{Fix}^G(M) \neq 0.$$

Choose

$$0 \neq m \in \operatorname{Fix}^G(M).$$

Then $gm = m$ for all $g \in G$. Hence the one-dimensional subspace

$$km \subset M$$

is G-stable. Since M is irreducible, we must have

$$M = km.$$

Thus M is one-dimensional and G acts trivially on M. Therefore

$$M \cong k$$

as kG-modules, where k is the trivial representation.

Conversely, the trivial representation k is one-dimensional, hence irreducible. Therefore, up to isomorphism, the trivial representation is the unique irreducible kG-module.

■